

Adana İlindeki İnsanların İlaç Kullanım Alışkanlıkları

[Medicine Use Behaviors of People in the City of Adana, Turkey]

ÖZET

AMAÇ: Çalışmamızın amacı; Adana ilindeki insanların ilaç kullanım davranışlarını incelemektir.

YÖNTEM: Planlanan örnek büyüklüğü 1.222 idi. Hazırladığımız Akılcı İlaç Kullanım Anketi görüşmeyi kabul eden 18 yaş üstü 1.111 kişiye telefonla uygulanmıştır.

BULGULAR: Katılanların %57,2'si doktora danışmadan ilaç kullanmaktadırlar. Doktora danışmadan ilaç kullananların %98,4'ü ağrı kesicileri kullandığını söylemiştir. Katılımcıların %30,5'i grip ve soğuk algınlığı durumlarında doktora sormadan antibiyotik kullandıklarını, %47,9'u doktorun verdiği antibiyotikleri bitirmeden bıraktıklarını söylemişlerdir, %85,8'i ilaçların son kullanma tarihlerine baktıklarını söylemişlerdir.

SONUÇ: Çalışmamıza katılan kişilerin Akılcı İlaç Kullanım Skoruna göre puanlarını hesapladık. 100 üzerinden tüm bireylerin aldıkları ortalama puan 68,3±13,5 bulunmuştur. Bu da Adana ilindeki insanların akılcı ilaç kullanım bilgisinin yeterli olmadığını, halkın eğitimi ile bilinç düzeyinin artırılması gerektiğini düşündürmektedir.

SUMMARY

AIM: The purpose of our study is to examine the medicine use behaviors of people in Adana.

METHOD: The planned sample size was 1,222. The "Rational Use of Drugs Questionnaire" prepared by us was applied to 1,111 voluntary subjects over 18 years old telephone survey.

RESULTS: 57.2% of these participants said that they used drugs without consulting a doctor. 98.4% of these participants told that they used analgesics without consulting a doctor. 30.5% said that they used antibiotics without asking a doctor in cases of common cold; 47.9% told that they discontinued the antibiotics the doctor prescribed for them before completely finishing; 85.8% said that they checked the expiry date of the medicines.

CONCLUSION: The average Rational Use of Drug Score of all the participants is found to be 68.3±13.5 on the scale of 100. This score shows that the knowledge of the people in Adana about rational use of drug is not sufficient and that education and awareness levels of the people should be raised.

Neslihan Pınar¹
Yusuf Karataş²
Nafiz Bozdemir³
İlker Ünal⁴

¹Department of Pharmacology, Faculty of Medicine, Inonu University, Malatya.

²Department of Pharmacology, Faculty of Medicine, Çukurova University, Adana.

³Department of Family Practice, Faculty of Medicine, Çukurova University, Adana.

⁴Department of Biostatistic, Faculty of Medicine, Çukurova University, Adana.

Anahtar Kelimeler:

Adana, Akılcı İlaç Kullanımı, Akılcı İlaç Kullanım Skoru, Telefonla Anket.

Key Words:

Adana, Rational Use of Drug, Rational Use of Drug Score, Telephone Survey.

Sorumlu yazar/

Corresponding author:

Neslihan PINAR
Department of Pharmacology, Faculty of Medicine, Inonu University, Malatya, Turkey.
neslihan.pinar@inonu.edu.tr

Gönderme Tarihi/Date of Submission: 23.08.2012, Kabul Tarihi/Date of Acceptance: 26.02.2013, DOI:10.5455/pmb.1-1344333007

INTRODUCTION

World Health Organization defined rational use of drug as "Rational use of drugs requires that patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements for an adequate period of time, and the lowest cost to them and their community".

The process of Rational Drug Use (RDU) is a systematic approach including careful and correct diagnosis of the patient's condition, determination of the treatment purposes, selection of an efficacy-

proved and reliable treatment among other various options, prescribing in accordance with this selected treatment way selected treatment way, starting the treatment after giving clear information and advices to the relevant patient, and finally monitoring and evaluating the results of the treatment (1).

Drugs have a very important place in human and public health, since they can bring the negative effects threatening human life and health to an end when used properly, on the other hand they can cause deaths when used improperly (2).

As some of the undesired effects of irrational drug use, we can give these following examples: adverse reactions and drug toxicity resulting from wrong drug selection; uncompleted treatments of the diseases; resistance development due to antibiotic usage at inadequate dosages and/or for inadequate period of time; high treatment expenses arising from deliberate use of expensive drugs in unnecessary cases. Misadministration of drugs, improper use of new and expensive products, misuse of antibiotics and injectable forms are the most commonly observed examples of irrational drug use in the countries where polypharmacy starts to exist (3).

Irrational drug use results from interrelated causes such as socio-cultural, economical and educational problems and administrative and regulatory mechanisms. Patients, doctors and other medical personnel, relevant authorities, and drug industry are mainly responsible for this problem (4).

Success of pharmacotherapy substantially depends on patients' compliance. Accurate drug selection and as well as accurate and legible prescription do not ensure and guarantee patients' proper use of the medicines. One of the main principles of rational drug use is that patients should be informed about pharmacological effects and side effects of the drugs and their possible interactions with other drugs and they should also be informed about both how to use the drugs and how to follow the instructions about the drugs (1).

Deterioration of one's health and observation of side-effects are the unavoidable consequences of irrational drug use. Improper use of analgesics and psychotropic drugs in children and adolescents is potentially dangerous to health, since these drugs have toxic and harmful side-effects. It is found that analgesics are widely used among adolescents (5).

The fact that many families have inadequate knowledge about drugs and store them in an insecure way leads to improper drug use in children and adults. Most improperly used drugs in children and adults are found to be analgesics (6). Over the counter analgesics are the most widely used drugs. Furthermore, it is shown that especially between the ages of 25 and 44, these drugs are more widely used in case of stress-related diseases, shoulder, neck and

arm pains, headaches and symptoms of common cold (7).

One of the most commonly observed way of irrational drug use over the world is self-medication with antibiotics. While using antibiotics, some criteria should be fulfilled such as right patient, accurate dosage level, accurate administration way, and right time. Inappropriate antibiotic use for the treatment of infections constitutes problems in terms of both treatment costs in all over the world and development of resistance to bacteria (8-10).

In 2010, the cost of drug consumption per person was found to be 133 Dollars in Turkey, whereas that value was found to be 956 Dollars in USA. This shows us that the level of drug consumption per person is higher in developed countries. Antibiotics constitute the biggest proportion of drug costs and they are followed by cardiovascular system drugs, antirheumatic drugs, nervous system drugs, oncologic drugs, respiratory system drugs. The proportion of antibiotics in the total drug costs was 19.9% while this percentage decreased to 13.9% in 2010. The costs of cardiovascular system drugs decreased from 14.1% to 11.4%. In addition, oncologic drugs constituted 3.8% of the total drug costs while they constituted 7.2% in 2010 (11).

When people become ill, they tend to use any of their existing drugs, use drugs upon the advice of their relatives, friends, etc. or use some certain drugs they directly bought from a pharmacy. By this way, self-medication (which is defined as drug use according to person's own knowledge or advices of non-professionals), in other words, habit of improper drug use emerges and this can cause dangerous situations (12).

This is the first study in that it aims to define the drug use habits of people in the city of Adana. Moreover this study also aims to serve as a model for pharmacoepidemiologic studies, to examine the drug use behaviors of the people in that region, to determine the factors affecting these behaviors, and to form a database.

MATERIAL and METHOD

This study is an epidemiological research aimed at surveying the drug use habits of the people in Adana, Turkey. In addition, it is a cross-sectional and

descriptive epidemiological research aimed at determining the current situation.

Each house which has a landline from the operator of Türk Telekom formed the universe of our study. In our study, samples were selected with the method of "cluster sample". Some of the data about the samples used in our study are as follows: total telephone number is 492,987; estimated prevalence is 15%, margin of error is 1% and the study is within 95% confidence interval. In addition, calculated sample size is 1,222. Prevalence was calculated with the sample size formula of epi info ($n = N \cdot z^2 \cdot p \cdot (1-p) / d^2 \cdot (N-1) + z^2 \cdot p \cdot (1-p)$).

The telephone numbers to be called for our study were chosen from the telephone centrals of Türk Telekom in Adana. The number of landlines was proportionally chosen by regarding the total number of landlines in countries and villages.

Those interviews were made via landlines belonging to houses and offices. Mobile phone numbers were excluded from the scope of our study. The study was carried out by randomly dialing the chosen telephone numbers and interviewing with the first person above 18 who answers the phone.

After being selected, all the telephone numbers were entered into a computer. When clicking on these phone numbers, the questionnaire came up. The interviews were made via a landline in Telephone and Internet Site Research Unit of Çukurova University. Besides, these interviews were made between the hours of "10.00 am and 06.00pm". Start and end dates were "February 2009 and June 2009" respectively.

In our study a "Rational Usage of Drug Questionnaire", which contains 32 questions related to socio-demographic attributes, knowledge and manners of people with respect to rational drug use, and insensible consumption of over the counter drugs, was used. a reliability test was carried out by interviewing with 30 people on the phone in order to check whether the questions in the survey were clear and understandable and as well as to control whether the desired results can be achieved or not. The same questionnaire was re-applied to the same people 15 days after the initial one. Accordingly, the questions with low reliability were excluded. In the analysis of the reliability test, the questions were found to be 84% reliable in general.

The last version of this questionnaire (containing 32 questions) was saved in a computer by Telephone and Internet Field Research Unit of Çukurova University. The first seven questions were concerned with the socio-demographic attributes (gender, year of birth, educational status, occupation, social

security availability and settlement). As of the 8th question, there are questions concerning with drug use without consulting a doctor, drug use upon the advice of non-professionals, advising medicines, the types of medicines that are used, use of antibiotic without consulting a doctor, discontinuation of the antibiotics before finishing completely and the reasons of it, the ways they use and store the medicines, use of herbal medicines, and conditions of the unused medicines.

At the beginning of each interview, the subject was informed about the survey and his/her consent was taken. The people who did not give consent were registered as "did not want to make an interview". Those interviews lasted approximately "3 to 3.5 minutes".

Determination of Rational Drug Use Scores

While determining the rational drug use scores, the questions which relate to rational drug use knowledge of the public and behaviors of the public with respect to rational drug use, were taken into account. The questions were divided into three groups namely Group A, Group B and Group C according to the order of their importance. Each question in Group A, Group B and Group C is equivalent to 2, 4, and 6 points, respectively.

According to this, the total score that a subject can get was calculated as 56. And then these scores were calculated on a scale out of 100. For example; if a participant gets 42 points out of 56, it corresponds to 75 ($42 \cdot 100 / 56 = 75$) on the scale of 100.

Statistical Analyses

SPSS 17.0 package program was used for the statistical analysis of the data. While categorical measurements (like gender and educational background) were expressed as numbers and percentages, continuous measurements (such as age and information scores) were expressed as mean and standard deviation (as minimum, medium and maximum when required). Chi-Square test statistic was used to compare categorical measurements. On the other hand t test was used to compare the continuous measurements of two groups and one-way analysis of variance was used to compare those of more than two groups. In all tests, statistical significance level was taken as 0.05.

RESULTS

In the study we carried out, the total number of phone calls made to interview with the subjects was

1,222 and the total number of completed interviews was 1,111. We were planning to reach 1,222 numbers of people; however we could reach 1,111. That is in part because some of the phone numbers we obtained from Türk Telekom were no longer in service, and because we could not get into contact with some of them. As a result, we managed to get into contact with 90% of the people we planned and we considered this rate as sufficient.

57.2% of the people (n = 635) we interviewed have said that they use a drug before consulting a doctor. 56.5% of the females and 58.3% of the males

have said they use a drug without consulting a doctor. When we compared these results according to the gender, no significant difference was found between females and males ($p > 0.05$) (Table 1).

51.6% of the illiterates, 45.8% of the literates, 52.9% of the primary school graduates, 58.1% of the secondary school graduates, 62% of the high school graduates, 62.4% of the university graduates have said that they use drug without consulting a doctor. Significant difference was found between different educational backgrounds ($p < 0.05$) (Table 1).

Table 1: Sociodemographic attributes of the participants.

	n	%
Gender		
Female	706	63.5
Male	405	36.5
Educational status		
Illiterate	93	8.4
Literate	24	2.2
Primary school graduate	508	45.8
High school graduate	292	26.3
University graduate	194	17.5
Age groups		
<25	146	13.2
25-55	721	65.0
55-65	161	14.5
>65	83	7.5
Social security		
Present	172	15.5
Absent	939	84.5

n: number of people in the group, %: The percentage of people from this group who use drug without consulting a doctor. (A significant difference was found between different educational backgrounds $p < 0.05$. A significant difference was found between city center and county $p < 0.05$).

64% of the ones with no social security and 55.9% of the ones with social security have told that they use drugs without consulting a doctor. No significant difference was found between them ($p > 0.05$). 60.8% of the ones living in city center and 56.6% of the ones living in county have told that they use drugs without consulting a doctor. And this difference was significant ($p < 0.05$) (Table 1).

Out of 1,111 people we interviewed, 635 (57.1%) have said that they use drugs without consulting a doctor; 14.9% have said that they advise drugs to their relatives, friends and neighbors; 7.9% have said that they use drugs upon the advices of their relatives, friends and neighbors (Table 2). Among 625 people (out of 635) who use drug without consulting a

doctor; 98.4% use analgesics, 53.1% use common cold medications, 34.3% use antibiotics, 18.3% use other drugs (Figure 1).

88 people out of 1,111 people we interviewed in Adana have said that they use drug upon the advices of their relatives, friends and neighbors. Out of these 88 people, 83 (94.3%) have said they use analgesics, 44 (50%) have said they use common cold medications, 26 (29.5%) have said they use antibiotics, 13 have said (14.8%) they use antiulcer drugs, 5 (5.7%) have said they use dermatological preparations, 2 people (2.3%) have said they use vitamins, 2 (2.3%) have said they use myorelaxants upon advise (Figure 2). (Some people mentioned more than one drug name).

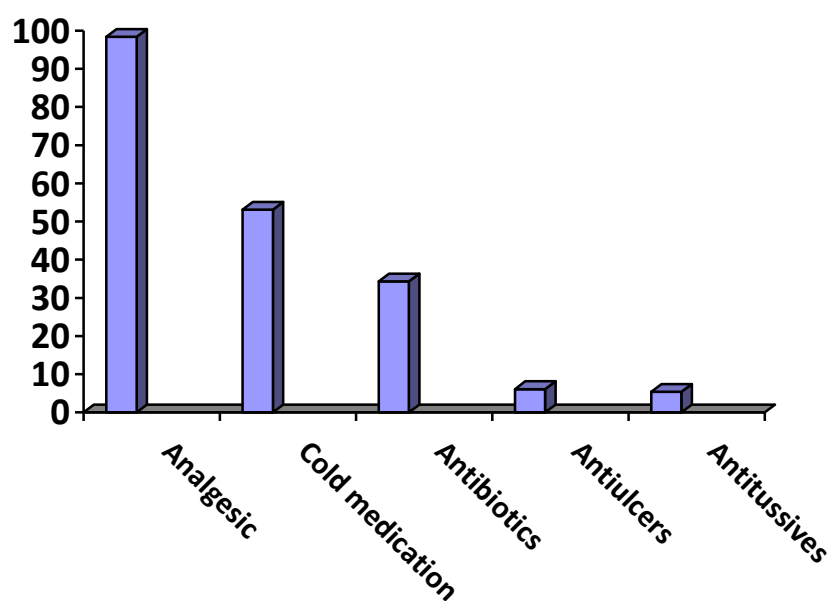


Figure 1. The percent of drugs used without consulting a doctor. (Analgesics are 98.4%, cold medications are 53.1%, antibiotics are 34.3%, other drugs are 18.3%).

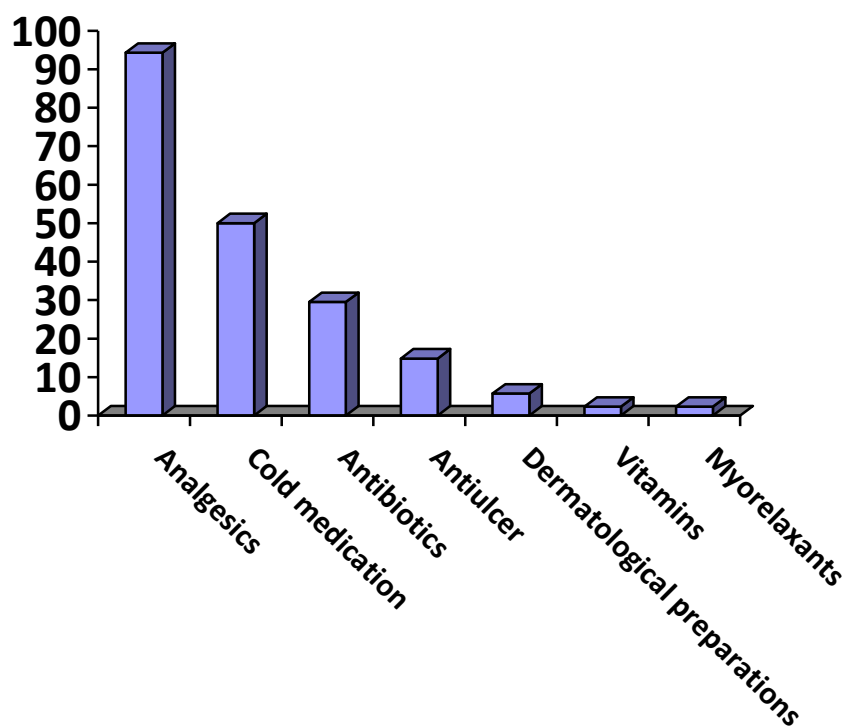


Figure 2. The percent of medicines used with relative, friend and neighbor advices. (Analgesics are 94.3%, cold medication are 50%, antibiotics are 29.5%, antiulcer drugs are 14.8%, other drugs are 10.3%).

Table 2: The ones who use drug without consulting a doctor

		n	Yes(%)
Gender	Female	706	56.5
	Male	405	58.3
Educational status	Illiterate		
	Literate	93	51.6
	Primary school graduate	24	45.8
	Secondary school graduate	403	52.9
	High school graduate	105	58.1
	University graduate	292	62.0
		194	62.4
Social security availability	Yes	172	55.9
	No	939	64.0
Settlement	City center	819	60.8
	County,village	292	43.4

n: number of people in the group, %: The percentage of people from this group who use drug without consulting a doctor.

(A significant difference was found between different educational backgrounds $p < 0.05$. A significant difference was found between city center and county $p < 0.05$).

When we have a look at the information of the participants regarding drug use, 339 people (30.5%) have stated they use antibiotics without consulting a doctor, 532 (47.9%) have stated they discontinue the antibiotics prescribed by the doctor, 361 (32.5%) have stated they ask for information on the drug from the doctor, 799 (72%) have stated they read prospectus of the drug they use, 995 (89.6%) have said they use the drug in the way prescribed by the doctor. From the participants, 953 (85.8%) have said they check the expiry date before using the drug, 321 (28.9%) have said they have a drug prescribed to keep at home, 549 (49.4 %) have said they apply herbal medicine at home (Table 3).

From the 532 people who discontinue the antibiotics prescribed by the doctor, 311 (58.4%) have said that they discontinue because of healing, 137 (25.7%) have said they discontinue because they feel well, 53 (9.9%) have said they discontinue because they do not want to use more drugs, 31 (6%) have said they discontinue because of other reasons (Table 4).

When side-effect is experienced while using the drug, out of 1111 people 451 (40.6%) have said they consult a doctor, 324 (29.2%) have said they have never experienced any side-effect, 195 (17.6%) have said they discontinue the drug, 97 (8.7%) have said they discontinue the drug and consult a doctor. 32 (2.9%) have said they continue the drug, 5 (0.5%) have said they ask about the side-effect to the pharmacist, 4 (0.4%) have said they look at the prospectus in case of side-effect and 3 (0.3%) have said they cure the side-effect at home with their own methods (Table 5).

When we have questioned the condition of unused drugs, 49% of the participants have said they throw away the drugs expired, 28% have said they do not keep unused drugs at home, 17.9% have said they keep the unused drugs at home, 13% have said they give them to a healthcare organization, 5.6% have said they give them to someone who need, 0.3% have said they use them for other purposes (Table 6).

Table 3: Information on the drug use of the participants

	n	%
The ones who use antibiotics without consulting a doctor	339	30.5
The ones who discontinue the antibiotics prescribed by the doctor	532	47.9
The ones who ask for information from the doctor on the drug prescribed by the doctor	361	32.5
The ones who read the prospectus of the drug	799	72.0
The ones who use the drug in the way prescribed by the doctor	995	89.6
The ones who check the expiry date before using the drug	953	85.8
The ones who have a drug prescribed to keep at home	321	28.9
The ones who apply herbal medicine at home	549	49.4

n: number of people in the group, %: The percentage of people

Rational drug use score

When we looked at the point distribution of rational drug use of the people we interviewed in city of Adana, the lowest point was found as 12.5 and the highest as 94.6. The average points of the individuals were found 68.3 ± 13.5 . We see that the highest point distribution spot is between 60 and 80. The highest scorers were found between 74 and 76.

When we looked at the distribution according to educational backgrounds, the average points of illiterates were found as 60 ± 16 , the lowest point 12.5 and the highest point 91. The average points of literates were found as 63.6 ± 15 , the lowest point 26.9 and the highest point 85.7. The average points of primary school graduates were found as 68.9 ± 13 , the lowest point 21 and the highest point 94. The average points of secondary school graduates were found as 71.5 ± 10 , the lowest point 40 and the highest point 94. The average points of high school graduates were found as 68.4 ± 13 , the lowest point 25 and the highest point 94. The average points of university graduates were found as 68.8 ± 12 , the lowest point 35 and the highest point 94 (Figure 3).

Table 4: The reasons for discontinuing the antibiotics prescribed by the doctor

	n	%
Healing	311	58.4
Feeling well	137	25.7
Not to use more drugs	53	9.9
Other reasons	31	6.0

n: number of people in the group, %: The percentage of people

Table 5: The things done in case a side-effect is experienced while using a drug

	n	%
I consult a doctor	451	40.6
I discontinue the drug	195	17.6
I discontinue the drug and consult a doctor	97	8.7
I continue to use the drug	32	2.9
I ask the pharmacist	5	0.5
I look at the prospectus	4	0.4
I cure it at home	3	0.3

n: number of people in the group, %: The percentage of people

Table 6: The condition of unused drugs

	n	%
I throw it away if expired	544	49.0
I do not keep drugs at home	317	28.5
I keep it at home	199	17.9
I give it to a healthcare organization	152	13.7
I give it to someone who need	62	5.6
Other reasons	3	0.3

n: number of people in the group, %: The percentage of people

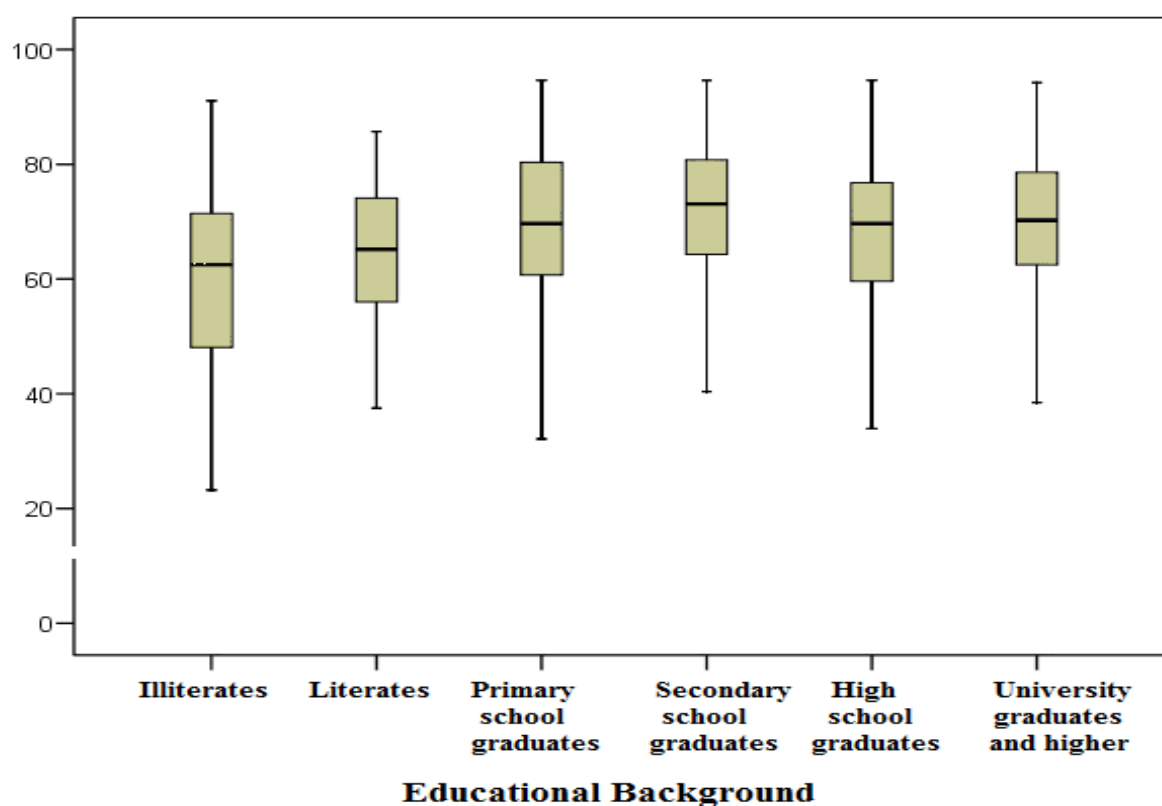


Figure 3. Distributions of rational drug use score points according to educational background (One-way analysis of variance was carried out). A significant difference was found between average points of illiterates and primary school graduates, secondary school graduates, high school graduates and university graduates ($p < 0.05$). No significant difference was found between the other groups ($p > 0.05$). Bonferroni was used)

DISCUSSION

Our study is an epidemiological research aimed at surveying the rational drug use behaviors of the people in Adana. In addition, it is a cross-sectional and descriptive epidemiological research aimed at determining the current situation.

When we examined the educational background, the highest rate was 36.3% in primary school graduates. 84.5% of the participants have stated that they have social security and 73.7% have stated that they live in city center. This shows that most of the participants can access to doctors and drugs conveniently.

57.7% of the people in our study have stated that they use drugs without consulting a doctor. In line with our study, 40-60% of the people interviewed in a study in Vietnam stated that they used drugs without consulting a doctor (13). In a study conducted in Slovenia, it was found that 77.2% of the participants used drugs without consulting a doctor when their symptoms started to appear (14).

Although there was no significant difference according to gender between the people stating they use drugs without consulting a doctor, a significant difference was found according to their educational background. On the other hand, in a study carried out in Denmark, it was found that drug use without consulting a doctor in headache was higher in females than the males (15). In a study conducted on women in Turkey, rational drug use habits of women were found to be inadequate (16).

45.8% of the literates and 62.4% of university graduates have told they use drugs without consulting a doctor. We can understand from this finding that as education increases, drug use rate without consulting a doctor also increases. As the education level increases, the access to medicine gets easier. In our study, most of the subjects with lower education level live in counties and villages. Thus, access of these people to medicine is more difficult than that of those living in cities. Because of financial impossibility, which causes trouble for people to go to doctor, people with higher education and financial possibility can more readily access to medicine.

Drug use rate without consulting a doctor was found 57.1%, rate of drug use upon the advice of relatives, friends was found 14.9% and rate of the ones advising drugs was found 7.9%. Although people behave more timidly in advising drugs for others, we see that they can use drugs more easily on

their own. The drugs which are used the most without consulting a doctor were found to be analgesics with 98.4%, followed by common cold medications with 53.1%, antibiotics with 34.3% and other drugs with 18.3%. In the study carried out with 6500 people in Finland, the rate of people who use prescribed analgesics a few times in a week or daily were found 8.7% and rate of people who use unprescribed analgesics were found 8.8%¹⁷. Moreover, in a study conducted in Nigeria, the most used drugs without consulting a doctor were found to be analgesics with 31.4%, followed by vitamins with 17.7%, antibiotics with 11.5% (18). Furthermore, in a study carried out in Palestine analgesics and antibiotics were found to be 86.6% and 20%, respectively (19). In a study performed in Belgium, 42.6% of the participants stated that they regularly used analgesics in case of headaches without consulting a doctor (20).

Antibiotic use without consulting a doctor was found 30.5% in our study. This rate was found 83.1% in the study carried out by Buke et al. (21). It was found that the people without social security used more antibiotics without consulting a doctor than the ones with social security in case of flu or common cold ($p < 0.05$).

48% of the people interviewed stated that they discontinued the antibiotics prescribed by the doctor without completely finishing. 58.3% of the people who discontinue antibiotics stated that they discontinued it because they felt well. In the studies conducted in Turkey, 67% of people stated they discontinued antibiotics because they recovered, and in another study 25.8 % of the students stated they discontinued drugs when the symptoms of the illness disappeared (22). A study conducted in university hospitals showed that antibiotics were used irrationally (23,24). Another study demonstrated that resistance developed against antibiotics in relation with irrational drug use in hospitals (25). When the antibiotics are discontinued upon the thought of being recovered, microorganisms develop resistance against antibiotics due to inappropriate antibiotic use. A danger which is more important than the economic burden caused by wrong drug use is the increase of resistant microorganisms.

89.6% of people have stated they use the drugs as recommended by the doctor and 85.8% of them have stated they check the expiry date before using the drug. This rate being high is a pleasing situation for us, because the drugs may be expired when buying from a pharmacy and may even cause deaths.

40.6% of the people have told they consult a doctor when they see a side-effect when using a drug;

consulting a doctor when a side-effect is seen is a rational approach. 49% of the people have told they throw away the expired drugs. A great amount of drugs are wasted in the world every year due to irrational drug use. A study carried out in Turkey in 2006 showed that 7% of the drugs in the pharmacies were thrown away when expired and 60% of the drugs expired without opening their boxes at homes. The cost of wasted drugs reaches up to about 500 million dollars (26).

The rate of people who have drugs prescribed to keep at home was found 28.9%. Some of the drugs kept additionally at home are wasted without being used. The people who keep drugs at home either use the drugs at home when they are sick or give them to someone who need. In a study in Switzerland, the money spent on drugs was found 1.6 billion euro; while the prescribed drugs constitute the 77% of the total expenses, prescribed drugs constitute the 84% of the drugs used in total (27).

In our study, 49.4% of the people have stated they apply herbal medicine at home. In the studies carried out with cancer patients, the most used alternative treatment method was found as herbal medicine (28,29).

The people participating in the research got 68.3 in average from Rational Drug Use Score which was calculated over 100. The lowest score was 12.5 and the highest score was 94. Since the average points obtained for city of Adana was over 50, it can be considered that rational drug use awareness of people is high. No difference was seen between the average points of males and females. It was shown that males and females got equal points in rational drug use. In the Rational Drug Use Score points, the highest point was obtained by primary school and higher graduates and the highest average point was obtained by secondary school graduates. We can understand from these values that RDUS point of educated people increases.

The attitudes of the society when they are sick are affected by many factors such as perception of health, health knowledge levels, educational background, income levels, expectations from healthcare organizations. Therefore, different regions give different results. In determination of drug use attitudes and behaviors of people, sociodemographic attributes of patients and influence of social environments on them as well as the drug knowledge of the physicians with the authority to prescribe drugs, their approach to the patient and instruction of the patients play a role (30, 31).

ACKNOWLEDGEMENTS

This study was supported by Çukurova University TF2009LTP8 research project. The experimental protocols were confirmed by the Local Ethics Committee of the Çukurova University Medical Faculty non Interventional Clinical Research. It was presented in the 21st National Pharmacology Congress in Eskişehir, Turkey 2011.

REFERENCES

1. Akıcı A, Kalaça S, Uğurlu MU, Toklu HZ, İskender E, Oktay Ş. Patient Knowledge About Drugs Prescribed at Primary Healthcare Facilities. *Pharmacoepidemiology and Drug Safety* 2004; 13: 871-6.
2. Phillips DP, Breder CC. *Annu. Rev. Public Health* 2002; 23: 135-150.
3. Siddigi S, Hamid S, Rafique G, Chaudry SA, Ali N, Shahab S, et al. Prescription Practices of Public and Private Health Care Providers in Attock District of Pakistan. *Internal Journal of Health Planning and Management* 2002; 17: 23-40.
4. Akıcı A, Kalaça S, Uğurlu MU, Karaalp A, Çali Ş, Oktay Ş. Impact of a short postgraduate course in rational pharmacotherapy for general practitioners. *BR. J. Clin. Pharmacology* 2004; 57: 310-21.
5. Holstein BE, Hansen EH, Due P. Social Class Variation in Medicine Use Among Adolescents. *Eur J Public Health* 2004; 14: 49-52.
6. Holstein BE, Andersen A, Krolner R, Due P, Hansen EH. Young Adolescents Use of Medicine for Headache: Sources of Supply, Availability and Accessibility at Home. *Pharmacoepidemiology and Drug Safety* 2008; 17: 406-10.
7. Koushede V, Holstein BE, Andersen A, Ekholm O, Hansen EH. Use of Over-the-Counter Analgesics and Perceived Stress Among 25-44 Year Olds. *Pharmacoepidemiology and Drug Safety* 2010; 19: 351-7.
8. İlhan MN, Durukan E, İlhan SÖ, Aksakal FN, Özkan S, Bumin MA. Self-Medication with Antibiotics: Questionnaire Survey Among Primary Care Center Attendants. *Pharmacoepidemiology and Drug Safety* 2009; 18: 1150-7.

9. Li LJ, Wang PS. Self-Medication with Antibiotics: a Possible Cause of Bacterial Resistance. *Med Hypotheses* 2005; 65:1000-1.
10. Shankar PR, Partha P, Shenoy N. Self-Medication and non-Doctor Prescription Practices in Pokhara Valley, Western Nepal: a Questionnaire-Based Study. *BMC Fam Pract* 2002;17: 17.
11. Pinar N. Ülkemizde İlaç Harcamaları. İnönü Üniversitesi Tıp Fakültesi Dergisi 2012; 19: 59-65.
12. Holstein BE, Hansen EH, Andersen A, Due P. Self-Rated Health as Predictor of Medicine Use in Adolescence. *Pharmacoepidemiology and Drug Safety* 2008; 17: 186-92.
13. Okumura J, Wakai S, Umenai T. Drug Utilisation and Self-Medication in Rural Communities in Vietnam. *Social Science & Medicine* 2002; 54: 1875-86.
14. Klemenc- Ketis Z, Kersnik J. The effect of demographic characteristics on self-medication patterns: a cross-sectional nationwide study from Slovenia. *Coll Antropol* 2011; 35: 1237-42.
15. Andersen A, Holstein BE, Due P, Hansen EH. Medicine Use for Headache in Adolescence Predicts Medicine Use for Headache in Young Adulthood. *Pharmacoepidemiology and Drug Safety* 2009; 18: 619-23.
16. Yılmaz M, Güler N, Güler G, Kocataş S. Bir Grup Kadının İlaç Kullanımı İle İlgili Bazı Davranışları: Akılcı mı? *Cumhuriyet Tıp Derg* 2011; 33: 266-77.
17. Turunen JHO, Mantyselka PT, Kumpusalo EA, Ahonen RS. Frequent Analgesic use at Population Level: Prevalence and Patterns of Use. *Pain* 2005; 5: 374-81.
18. Auta A, Omale S, Folorunsho TJ, David S, Banwat SB. Medicine Vendors: Self-medication Practices and Medicine Knowledge. *N Am J Med Sci* 2012; 4:24-8.
19. Sawalha AF. A descriptive study of self-medication practices among Palestinian medical and nonmedical university students. *Res Social Adm Pharm* 2008; 4:164-72.
20. Mehuys E, Paemeleire K, Van Hees T, Christiaens T, Van Bortel LM et al. Self-medication of regular headache: a community pharmacy-based survey. *Eur J Neurol* 2012;
21. Büke C, Limoncu MH, Emertcan S, Ciceklioglu M, Tuncel M, Köse T, Eren S. "Irrational Use of Antibiotics Among University Students. *Journal of Infection* 2005; 51: 135-9.
22. Bilgili N, Karatay G. Sait Yazıcı Sağlık Ocağı Bölgesinde Yaşayan Bireylerin İlaç Tüketimi İle İlgili Bazı Uygulamaların Belirlenmesi. *Hemşirelik Yüksekokulu Dergisi* 2005; 39-48.
23. Saba R, İnan D, Turhan Ö, Yalçın AN, Günseren F, Mamikoğlu L. Surveillance of antimicrobial use in a Turkish university hospital. *Turk J Med Sci* 2011; 41: 701-710.
24. Yılmaz GR, Bulut C, Yıldız F, Arslan S, Yetkin MA, Demiröz AP. Examining Antibiotic Use at an Education and Research Hospital in Turkey: Point Prevalence Results. *Turk J Med Sci* 2009; 39: 125-131.
25. İncecik Ş, Saltoğlu N, Yaman A, Karayaylalı İ, Özalevli M, Gündüz M, Burgut R. The Problem of Antimicrobial Resistance in Nosocomial Medical and Surgical Intensive Care Units Infections in a University Hospital; a Two-Year Prospective Study. *Turk J Med Sci* 2009; 39: 295-304.
26. Özata M, Aslan Ş, Mete M. Rasyonel İlaç Kullanımının Hasta Güvenliğine Etkileri: Hekimlerin Rasyonel İlaç Kullanımına Etki Eden Faktörlerin belirlenmesi, www.sosyalbil.selcuk.edu.tr/sos_mak/articles/2008/20/MO-SA-MM.PDF. 3.12.2009.
27. Wettermark B, Hammar N, Fored CM, Leimanis A, Otterblad Olausson P, Bergman U, et al. The new Swedish Prescribed Drug Register –Opportunities for Pharmacoepidemiological Research and Experience from the First Six Months. *Pharmacoepidemiology and Drug Safety* 2007; 16: 726-35.
28. Algier L, Hanoğlu Z, Özden G, Kara F. The Use of Complementary and Alternative (non-conventional) Medicine in Cancer Patients in Turkey. *European Journal of Oncology Nursing* 2005; 9: 138-46.
29. Aditya B, Debra LB, Larry JP, Brent AB, Timothy JM. Efficacy of Complementary and Alternative Medicine Therapies in Relieving Cancer Pain: A Systematic Review. *Journal of Clinical Oncology* 2006; 24: 5457-64.

TAF Preventive Medicine Bulletin, 2013: 12(6)

30. Mackenbach JP, Roskam AJ, Menvielle G. Socioeconomic Inequalities in Health in 22 European Countries. The New England Journal of Medicine 2008; 358: 2468-81.
31. Weitoft GR, Rosen M, Ericsson Ö, Ljung R. Education and drug use in Sweden – a nationwide register-based study. Pharmacoepidemiology and Drug Safety 2008; 17: 1020-8.

Copyright of TAF Preventive Medicine Bulletin is the property of GATA Halk Sagligi AD and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.